2.1 Passwords

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2.2 What is a Password?

A secret combination of alphanumerical characters and symbols used to verify identity and grant access to computer systems, online accounts and secured computer resources.

The idea is to protect personal information, data (such as personal identity, medical records, payment details etc...) and accounts from unauthorised access.

Secure practices when creating, storing, and using passwords. For example, the use of long, unique passwords for every new or existing account, and enabling MFA or 2FA to reduce the risk of unauthorised access.

Additionally, here are a list of elements it includes:

- Ensure the password is unique across all accounts
- Refrain from using predictable patterns (such as, names, dates etc)
- Use of complex, strong and long passwords (12-16+ characters)
- Enabling Multi-Factor Authentication (MFA) as mentioned above
- Update passwords when suspected of compromise
- Check breach databases for recognised passwords recorded in breaches Why Password Hygiene matters?

Poor password hygiene, weak passwords or reuse, unsafe password storage are currently one of the largest weaknesses in modern day cybersecurity. Attackers often make use of leaked credentials in stuffing attacks. Attackers will rely on the fact that people reuse the same password across various of different sites, platforms and services. Which is what Password Hygiene is built to prevent.

2.3 Benefits of Password Hygiene

2.3.1 Unique passwords

By using separate passwords across all platforms, sites and services you essentially eliminate the one advantage cyber attackers hope to have when executing their exploits (such as password stuffing). It greatly lowers the risk of unauthorised access.

Strong and long passwords

Using strong and long passwords makes it more difficult for attackers to guess or crack them. Short passwords and common words are easier to compromise, especially when attackers use automated tools to test thousands or millions of possible combinations.

A strong password should ideally be at least 12–16 characters long. Using a mixture of letters, numbers and symbols can help, but length and unpredictability are very important. Another good option is using a passphrase made up of several unrelated words, to make it unique and difficult to predict.

2.3.2 Multi-Factor Authentication (MFA)

Multi Factor Authentication adds another layer of security to an account. Instead of relying only on a password, the user must provide another form of verification, such as an authentication app, a security key or a code sent to their device.

This means that even if an attacker manages to obtain a password, they may still be unable to access the account without the additional verification method. Protection against data breaches

Websites and online services can experience data breaches, which may expose users personal information and login credentials. If a password has been exposed in a breach, attackers may attempt to use it to access other accounts belonging to the same person.

Good password hygiene helps reduce this risk by ensuring that passwords are unique and that compromised credentials are changed quickly. Users should also check whether their email addresses or passwords have appeared in known data breaches. Improved privacy and data protection

Passwords help protect sensitive information, including personal details, financial records, private messages and work-related documents. Maintaining good password hygiene reduces the chances of this information being accessed by unauthorised individuals.

This is especially important in workplaces where employees may have access to confidential or sensitive company information, customer records or internal computer systems.

2.4 Common Password Security Risks

2.4.1 Password reuse

Password reuse occurs when a person uses the same password for multiple accounts. This is risky because a single compromised account could allow an attacker to attempt access to several other services. (Like by using password stuffing)

For example, if someone uses the same password for their email, social media and online banking, a breach on one website could put the other accounts at risk. Using a different password for every account helps prevent this.

2.4.2 Weak and predictable passwords

Passwords such as "123456", "password" or a person's name are easy to guess. Attackers may also use personal information, such as birthdays, pet names or favourite teams, to make educated guesses.

Users should avoid information that can easily be discovered through social media or other public sources. Instead, passwords should be unpredictable and difficult for other people to associate with the account owner.

2.4.3 Phishing attacks

Phishing is a technique in which attackers pretend to be legitimate organisations or individuals to trick users into revealing sensitive information. They may send emails or text messages containing links to fake login pages.

If a user enters their password on one of these pages, the attacker may capture the credentials and use them to access the real account.

To reduce this risk, users should check website addresses carefully, avoid suspicious links and never enter passwords on websites they cannot verify as legitimate.

2.4.4 Unsafe password storage

Storing passwords in plain text documents, unprotected notes or easily accessible files can expose accounts if a device is lost, stolen or compromised.

A password manager is a useful alternative. It securely stores passwords and can generate strong, unique credentials for different accounts. Users should protect their password manager with a strong master password and enable MFA where available.
Best Practices for Password Hygiene

2.4.5 Use a password manager

A password manager helps users create, store and manage unique passwords without needing to memorise every single one. It can also help identify weak or reused passwords and generate stronger alternatives.

Users should choose a reputable password manager and protect access to it carefully.

2.4.6 Change passwords when necessary

Passwords do not always need to be changed at regular intervals if they remain strong and secure. However, they should be changed immediately if there is evidence that they have been compromised, exposed in a breach or accessed by an unauthorised person.

Users should also change default passwords on devices and systems, as these may be publicly known or easy to guess.

2.4.7 Keep recovery options secure

Account recovery methods, such as backup email addresses, phone numbers and recovery codes, should also be protected. If an attacker gains access to a recovery account, they may be able to reset passwords and take control of other accounts.

Recovery codes should be stored securely and should not be shared with other people.

2.4.8 Check for compromised passwords

Users can check whether their email addresses or passwords have appeared in known data breaches through reputable breach checking services. If a password has been exposed, it should be replaced with a new, unique one.

Users should never enter their current password into an unfamiliar website simply because it claims to check password security.

2.5 Conclusion

Password hygiene is an important part of cybersecurity because passwords are often the first line of defence against unauthorised access. Weak passwords, password reuse and unsafe storage can expose personal information and allow attackers to compromise multiple accounts.

By creating long and unique passwords, using a password manager, enabling Multifactor Authentication and responding quickly to suspected compromises, users can significantly improve their account security.

Ultimately, good password hygiene is not just about creating a strong password. It is about developing safe habits when creating, storing and using passwords to protect personal information, workplace systems and digital resources.

7 Password Hygiene Best Practices to Follow | Dashlane What is password stuffing?

What is credential stuffing? Meaning, examples, and prevention

I also utilised the AI overview for some of my research mainly for much simpler topics such as the risks of not using password hygiene.